Digital Forensics

Data Recovery Lab

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Task Overview

This lab simulated the accidental deletion of files from a storage device and their recovery using basic, freely available data recovery tools. Three real test files (a text file, a PDF and a JPEG) were placed on a FAT32-formatted USB stick, and three deletion scenarios were tested: a normal delete, a Shift+Delete, and a quick format of the whole stick. Each scenario was followed by recovery attempts with two tools, Recuva (a file-system-based recovery tool with a signature-scanning Deep Scan mode) and PhotoRec (a signature-based file carver from the TestDisk package). To make the results objective, the SHA-256 hash of every original file was recorded before deletion, and every recovered file was hashed and compared against it. All work was done on a dedicated USB stick, and all recovered files were saved to a different drive so that nothing was written to the stick between deletion and recovery.

Objectives

- Simulate the accidental deletion of files from a storage device in three scenarios: normal delete, Shift+Delete and quick format.
- Recover the deleted files from a FAT32 volume using two free recovery tools, Recuva and PhotoRec.
- Compare the two tools in terms of filename preservation, file size accuracy, speed, file types recovered and ease of use.
- Verify every recovery objectively using SHA-256 hashes taken before deletion.
- Explain, using how FAT32 stores and deletes files, why recovery works in some cases and fails in others.
- Document the whole process with screenshots, hash evidence and a written analysis of the results and limitations.

Learning Outcomes

By the end of this task, I should be able to:

- Practical understanding of how deletion works on FAT32: only the directory entry and the FAT are changed, so the file data remains on disk until the same clusters are reused.
- Hands-on experience recovering deleted files after a normal delete, a Shift+Delete and a quick format, and an understanding of why the results of the first two were the same on a removable drive.
- Applied knowledge of the difference between metadata-based recovery (Recuva standard scan) and signature-based carving (Recuva Deep Scan and PhotoRec), including why carving loses filenames and cannot identify plain text files.
- Experience verifying recovered data with SHA-256 hashes, including how to diagnose a hash mismatch (the 104 extra bytes on the Recuva-recovered JPEG) by hashing only the first bytes of the recovered file.
- Understanding of safe recovery practice: working on a dedicated device, saving recovered files to a different drive, and not writing to the affected device before recovery.
- Awareness of the factors that make recovery fail or become impractical, such as overwriting, fragmentation, full format or secure erase, encryption and SSD TRIM.

Theory

How File Deletion Works on FAT32

FAT32 keeps a directory entry for each file (holding its name, size and starting cluster) and a File Allocation Table (FAT) that records which clusters belong to which file. When a file is deleted, only this metadata changes: the first byte of the file's directory entry is set to a "deleted" marker, and the file's clusters are marked free in the FAT. The file's data stays on the disk until the same clusters are reused by new data. This is why Recuva showed deleted names with the first character replaced by an underscore, and why a deleted file can usually be recovered if nothing has been written to the device since. On removable drives Windows deletes files immediately rather than sending them to the Recycle Bin, so a normal delete and a Shift+Delete behave the same way.

Quick Format vs Full Format

A quick format rewrites the boot sector, the FAT and the root directory, but it does not erase the data area. Windows then shows the drive as empty, yet the file contents are still on the disk until they are overwritten. A full (non-quick) format is different, and in this lab it was used once before the tests to clear leftovers from the stick's earlier corrupted state so that they could not appear as false recoveries.

Recovery Tools: Recuva and PhotoRec

Recuva has two modes. The standard scan reads the file-system metadata and can list deleted files with their (partly damaged) names and a recoverability status. Deep Scan instead searches the disk for file signatures, which finds files that no longer have directory entries but gives them generic names. PhotoRec, part of the TestDisk package, ignores the file system entirely and carves files out of raw data by recognising their internal signatures (for example the header and end marker of a JPEG). Because it depends on signatures, it cannot identify plain text files, and it never restores the original filenames.

Hash Verification (SHA-256)

A cryptographic hash such as SHA-256 produces a fixed-length fingerprint of a file's contents. If a recovered file has the same SHA-256 hash as the original, the two files are byte-for-byte identical. Hashes were computed with the PowerShell Get-FileHash -Algorithm SHA256 command, recorded before any deletion, and compared with the hashes of the recovered files. A hash mismatch shows that the file differs in at least one byte, which then has to be investigated.

Data Recovery Lab: Recovering Deleted Files from a FAT32 USB Stick

1. Objective

To simulate the accidental deletion of files from a storage device and recover them using basic, freely available data recovery tools. The lab tests three deletion scenarios (normal delete, Shift+Delete, quick format) on a FAT32 volume, compares two recovery tools (Recuva and PhotoRec), and verifies every recovery with SHA-256 hashes taken before deletion.

2. Environment

Item	Detail
Host OS	Windows 11 Pro, version 10.0.26200 (64-bit)
Host hardware	Intel Core i5-8400, 16 GB RAM
Test device	16 GB USB flash drive (`\\.\PhysicalDrive3`, 15 GB / 14 GiB), previously corrupted, used only for this lab
File system tested	FAT32, volume label `LABFAT`, cluster size 8,192 bytes, 30,720,000 sectors
Tools	Recuva 1.55.0.133 (64-bit); PhotoRec 7.3-WIP (September 2025), part of the TestDisk package; both installed with `winget`
Hashing	PowerShell `Get-FileHash -Algorithm SHA256`
Safety measures	All work done on a dedicated USB stick, never on the system drive. All recovered files were saved to `C:\Recovered\...`, never to the

	USB stick. Nothing was written to the stick between deletion and recovery.

Test files

File	Type	Size (bytes)	SHA-256 (original)
note.txt	Plain text	54,018	[PASTE FROM SCREENSHOT 5]
doc.pdf	PDF (printed from Notepad)	71,134	[PASTE FROM SCREENSHOT 5]
photo.jpg	JPEG photograph	303,512	86D29B28FEE86ECA16000E5605D2AA09348B949531216631B7FDAC20FB206EBB

The originals were kept in `C:\LabSource`, and the hash list was saved in `C:\LabHashes\original_hashes.txt` (outside the source folder so it was not copied to the USB stick).

3. Methodology

1. **Prepare the device.** Full (non-quick) format of the USB stick to FAT32, so leftovers from the corrupted state could not appear as false recoveries.
2. **Create test data.** Three real files of different types (text, PDF, JPEG). Real files were used because PhotoRec identifies files by their internal signatures.
3. **Record baseline hashes.** SHA-256 of each original before any copying.
4. **Copy to the USB stick and verify.** Hashes of the copies on the stick matched the originals, which established a verified baseline.
5. **Delete in three scenarios.**
    - **A:** normal delete of `note.txt`
    - **B:** Shift+Delete of `doc.pdf`
    - **C:** quick format of the stick while `photo.jpg` was still on it (`doc.pdf` from Scenario B was still in unallocated space at that point)
6. **Recover with each tool** into folders on C:. Recuva was run in standard scan mode for A and B and in Deep Scan mode for C. PhotoRec was run with the *Other* file-system option and *Free* (unallocated space only) analysis.
7. **Verify.** Recovered files were hashed and compared with the original hashes, and each was opened in its normal viewer without editing or saving.

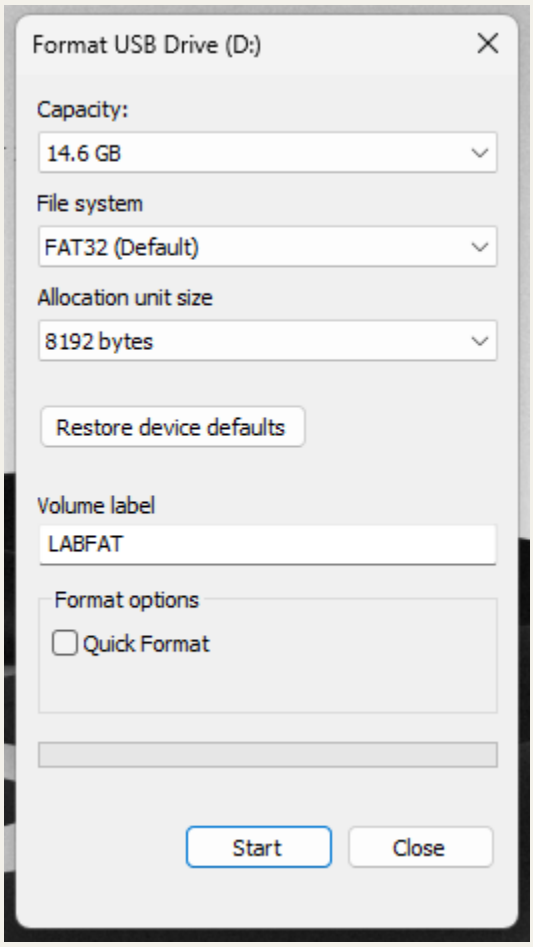
4. Steps with Screenshots

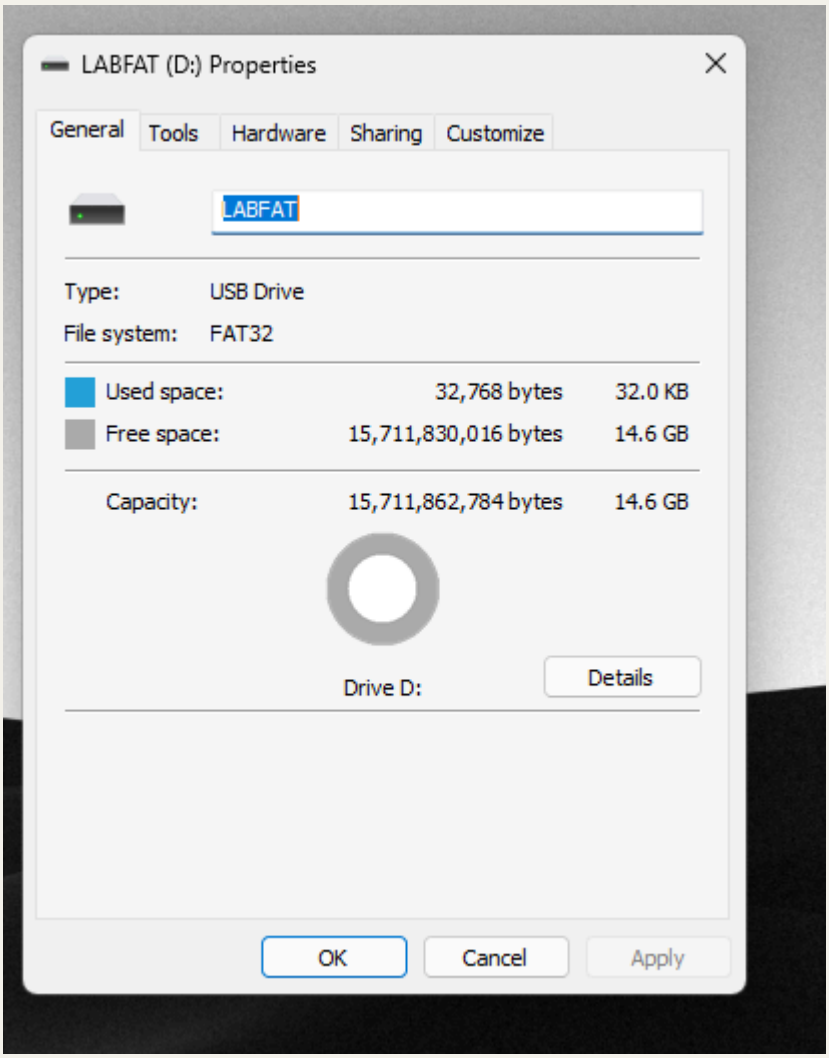
4.1 Identify and format the test device

- Identified the USB stick as `D:` (about 15 GB) and confirmed it was not the system drive.

Volume	Layout	Type	File System	Status	Capacity	Free Spa...	% Free
(D:)	Simple	Basic	RAW	Healthy (P...	14.65 GB	14.65 GB	100 %

- Full format to FAT32 (Quick Format unchecked), label `LABFAT`.




LABFAT (D:) Properties
General
Tools
Hardware
Sharing
Customize
LABFAT
Type: USB Drive
File system: FAT32
Used space: 32,768 bytes 32.0 KB
Free space: 15,711,830,016 bytes 14.6 GB
Capacity: 15,711,862,784 bytes 14.6 GB
Drive D:
Details
OK
Cancel
Apply

4.2 Create test data and baseline hashes

- Created `note.txt`, `doc.pdf` and `photo.jpg` in `C:\LabSource`.

```
PS C:\Users\newadmin\Desktop> dir C:\LabSource


    Directory: C:\LabSource


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         9/28/2026  10:04 PM          71134 doc.pdf
-a----         9/28/2026  10:03 PM          54018 note.txt
-a----        10/29/2025   1:23 PM         303512 photo.jpg
```

- Recorded SHA-256 hashes of the originals.

```
PS C:\Users\newadmin\Desktop> Get-FileHash C:\LabSource\* -Algorithm SHA256 | Select-Object Hash, @{n='File';e={Split-Path $_.Path -Leaf}} | Tee-Object -FilePath C:\LabHashes\original_hashes.txt | Format-Table -AutoSize

Hash                                                             File
----                                                             ----
7E1F76EAB5E60C7DD27B5785849E9569458341A70CF09DAFE85FFCB408F3D288 doc.pdf
8D4E600464D6E93703561BE946B8DF31503104F266175846FBFFAA9A3B846811 note.txt
86D29B28FEE86ECA16000E5605D2AA09348B949531216631B7FDAC20FB206EBB photo.jpg
```

- Copied the files to `D:` and confirmed the hashes on the stick matched.

```
PS C:\LabSource> Copy-Item C:\LabSource\* D:\
PS C:\LabSource> dir D:\


    Directory: D:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         9/28/2026  10:04 PM          71134 doc.pdf
-a----         9/28/2026  10:03 PM          54018 note.txt
-a----        10/29/2025   1:24 PM         303512 photo.jpg


PS C:\LabSource> Get-FileHash D:\* -Algorithm SHA256 | Select-Object Hash, @{n='File';e={Split-Path $_.Path -Leaf}} | Format-Table -AutoSize

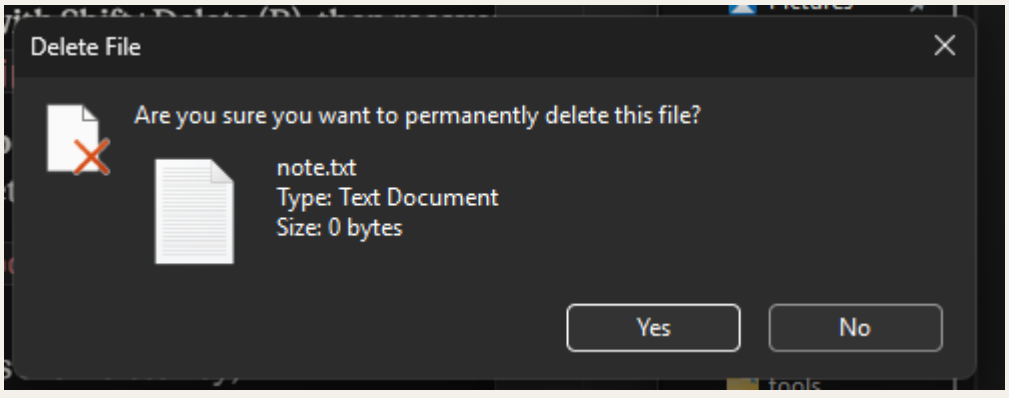
Hash                                                             File
----                                                             ----
7E1F76EAB5E60C7DD27B5785849E9569458341A70CF09DAFE85FFCB408F3D288 doc.pdf
8D4E600464D6E93703561BE946B8DF31503104F266175846FBFFAA9A3B846811 note.txt
86D29B28FEE86ECA16000E5605D2AA09348B949531216631B7FDAC20FB206EBB photo.jpg
```

4.3 Install tools

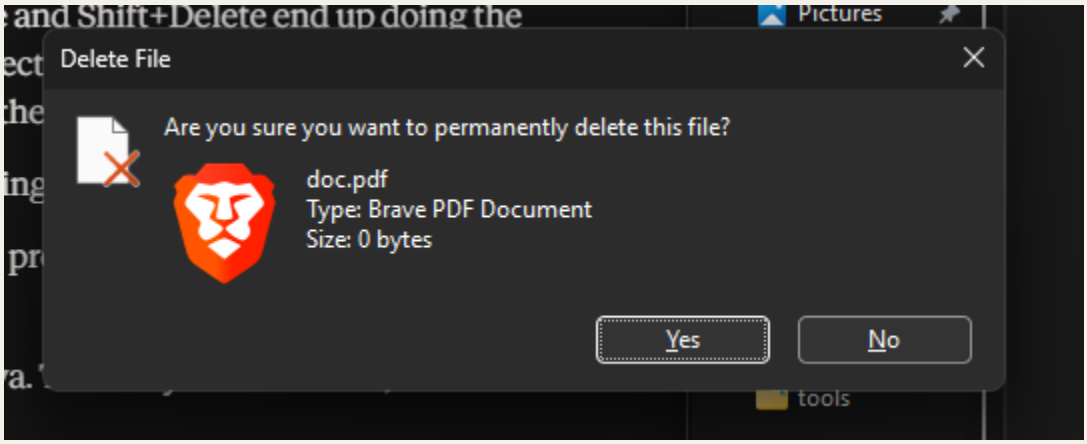
```
PS C:\LabSource> winget install --id Piriform.Recuva -e
Found Recuva [Piriform.Recuva] Version 1.55.0.133
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://download.ccleaner.com/rcsetup155.exe
  ██████████████████████████████  24.5 MB / 24.5 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\LabSource> winget install --id CGSecurity.TestDisk -e
Found TestDisk [CGSecurity.TestDisk] Version 7.3
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://www.cgsecurity.org/testdisk-7.3-WIP.win64.zip
  ██████████████████████████████  28.9 MB / 28.9 MB
Successfully verified installer hash
Extracting archive...
Successfully extracted archive
Starting package install...
Path environment variable modified; restart your shell to use the new value.
Command line alias added: "fidentify_win"
Command line alias added: "photorec_win"
Command line alias added: "qphotorec_win"
Command line alias added: "testdisk_win"
Successfully installed
PS C:\LabSource>
```

4.4 Scenario A: normal delete (note.txt)

- `D:` before deletion, then the delete prompt and the result. Windows showed a "permanently delete" prompt because removable drives do not use the Recycle Bin.

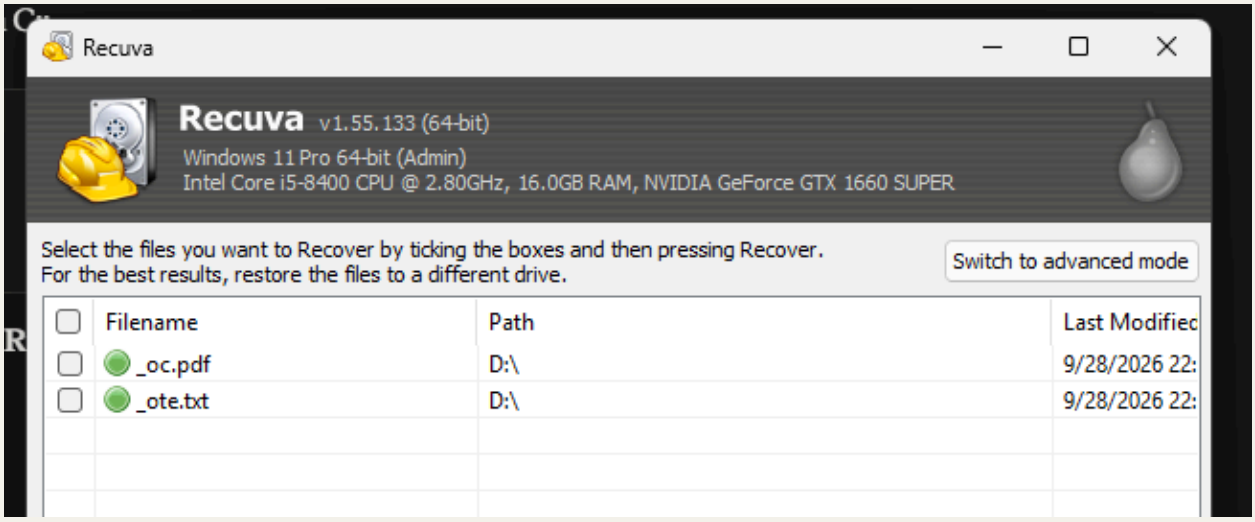


4.5 Scenario B: Shift+Delete (doc.pdf)

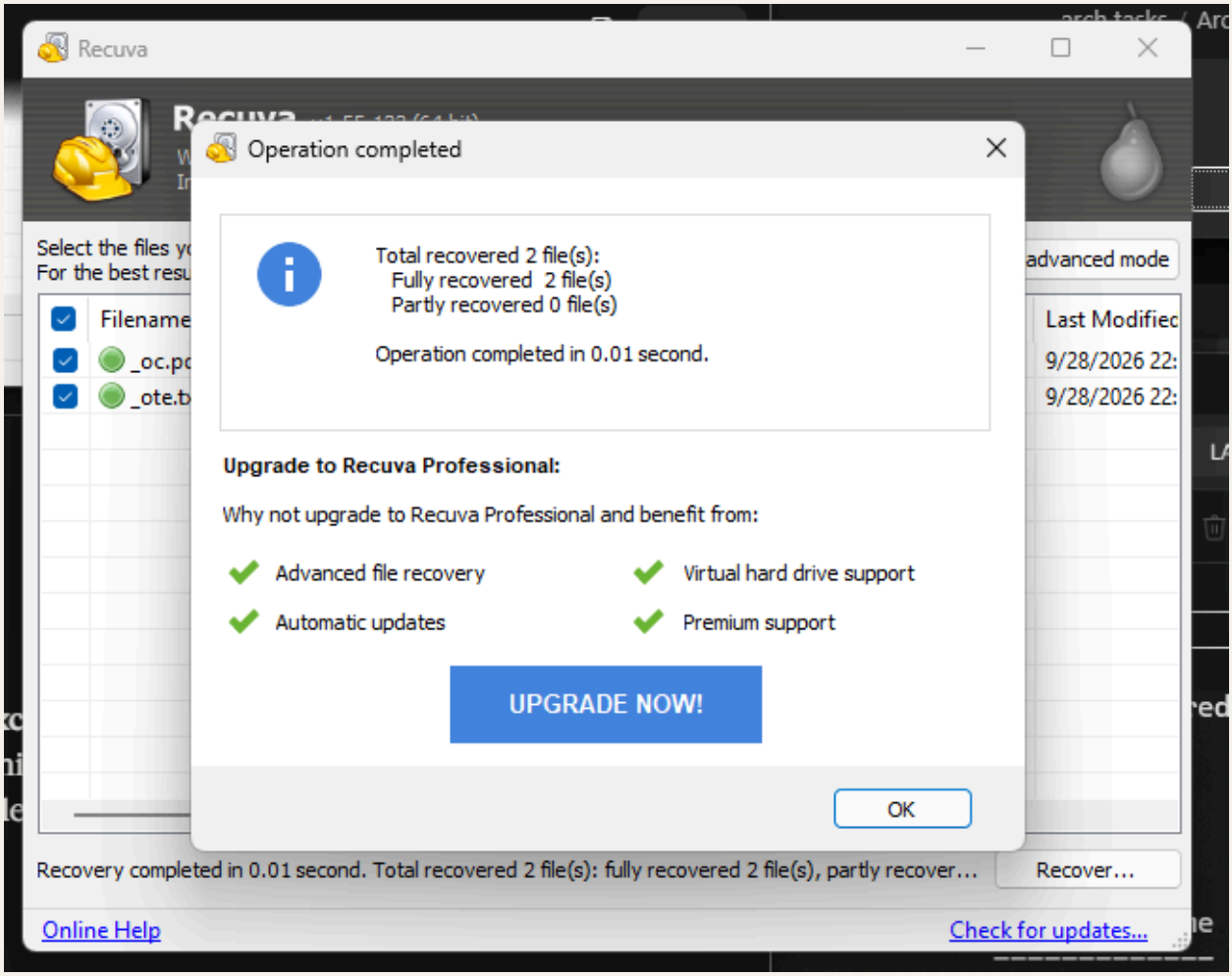


4.6 Recovery of A and B with Recuva

- Standard scan of `D:` listed both deleted files with green ("excellent") status. The first character of each name was replaced by `_` (`_ote.txt`, `_oc.pdf`). The scan took 0.02 seconds.



- Recovered both to `C:\Recovered\Recuva`.



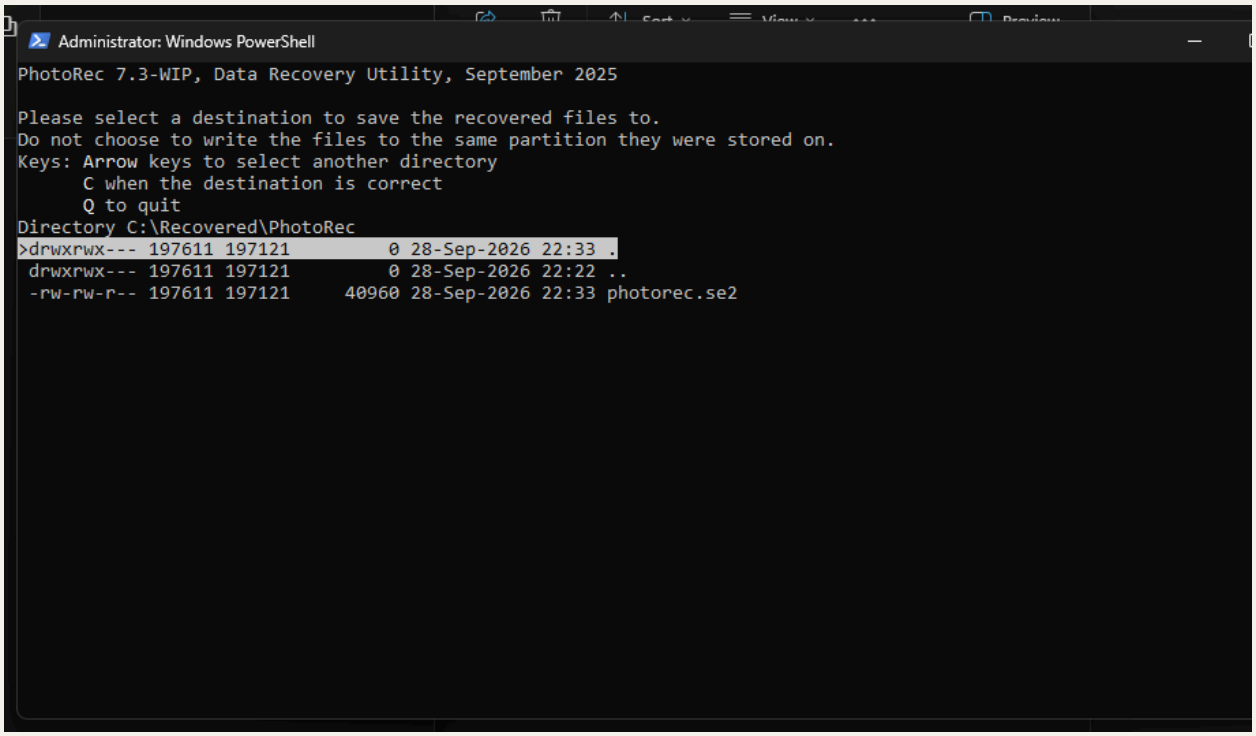
- Hash comparison: both matched their originals.

```
PS C:\LabSource> $o = Get-FileHash C:\LabSource\* -Algorithm SHA256
PS C:\LabSource> Get-FileHash C:\Recovered\Recuva\* -Algorithm SHA256 | ForEach-Object {
>>   $m = $o | Where-Object Hash -eq $_.Hash
>>   [pscustomobject]@{Recovered=(Split-Path $_.Path -Leaf); MatchesOriginal=[bool]$m; OriginalName=$(if($m){Split-Path
$m.Path -Leaf}else{'-'}); Hash=$_.Hash}
>> } | Format-Table -AutoSize

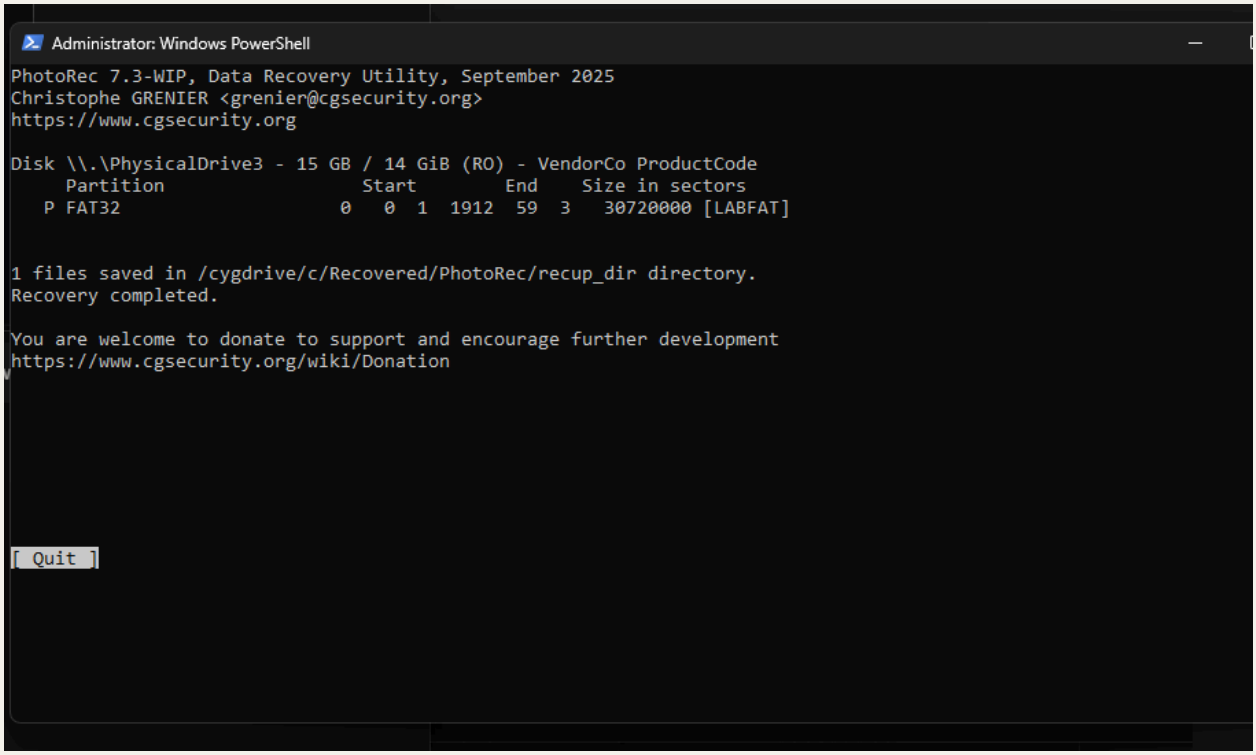
Recovered MatchesOriginal OriginalName Hash
--------- --------------- ------------ ----
_oc.pdf             True doc.pdf      7E1F76EAB5E60C7DD27B5785849E9569458341A70CF09DAFE85FFCB408F3D288
_ote.txt            True note.txt     8D4E600464D6E93703561BE946B8DF31503104F266175846FBFFAA9A3B846811
```

4.6b Recovery of A and B with PhotoRec

- Media: `\\.\PhysicalDrive3`, partition FAT32 `[LABFAT]`, file system type *Other*, analysis *Free*, output `C:\Recovered\PhotoRec`.

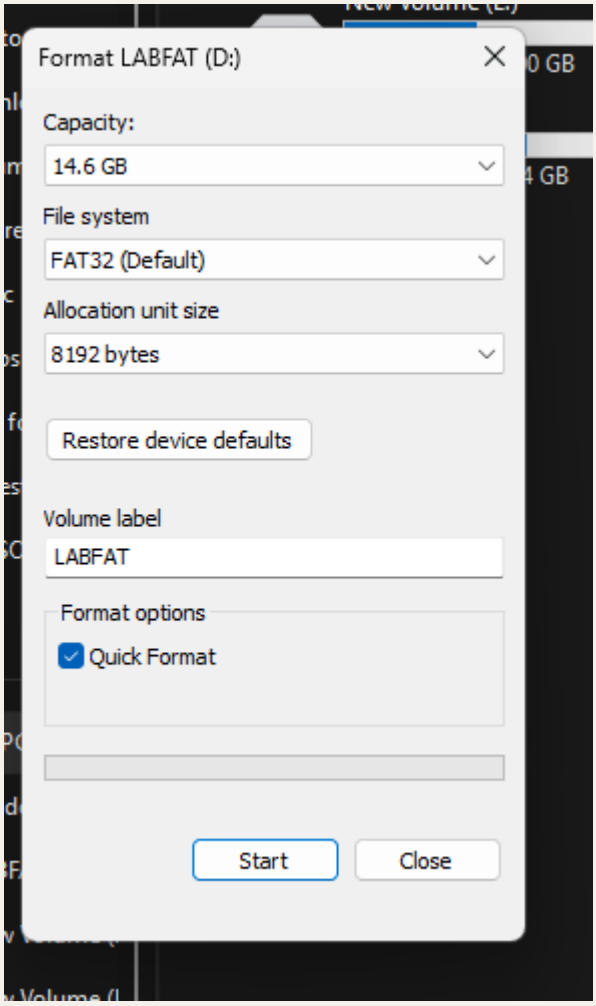


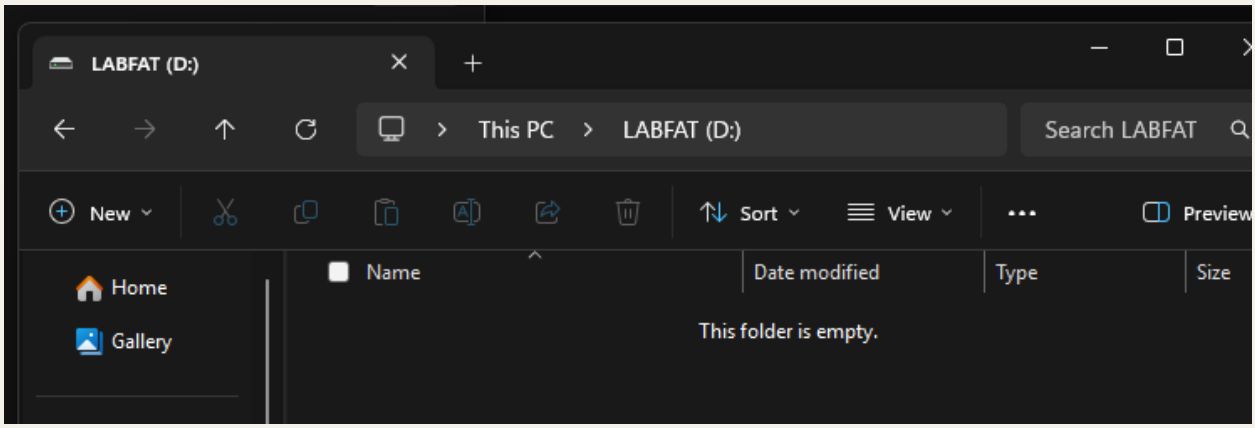
- PhotoRec saved one file: `f0032832_note_txt_Notepad.pdf`, hash-identical to `doc.pdf`. `note.txt` was not recovered.



4.7 Scenario C: quick format

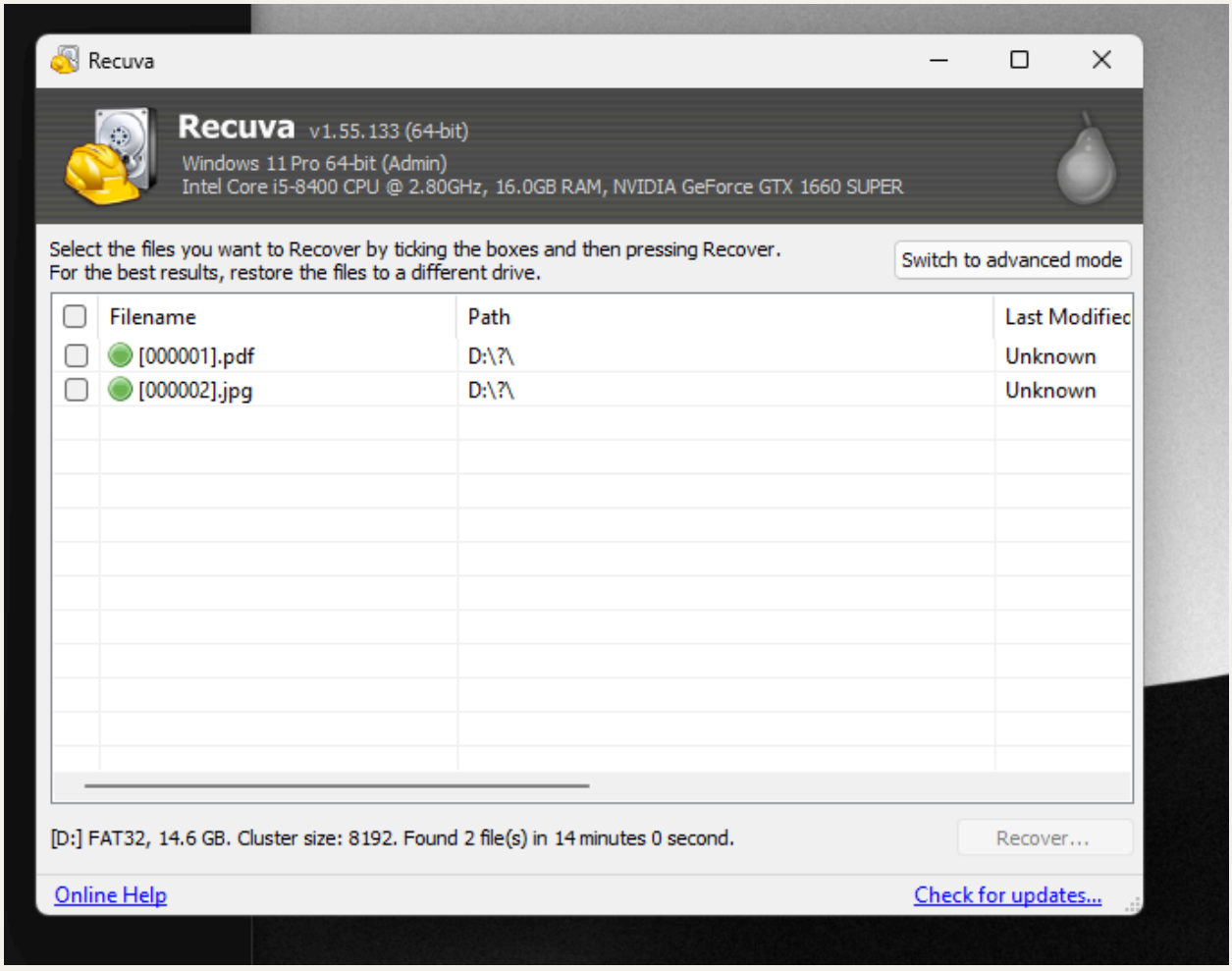
- Quick format of `D:` (FAT32, label `LABFAT`, Quick Format checked) with `photo.jpg` still on the stick. Windows then showed the drive as empty.



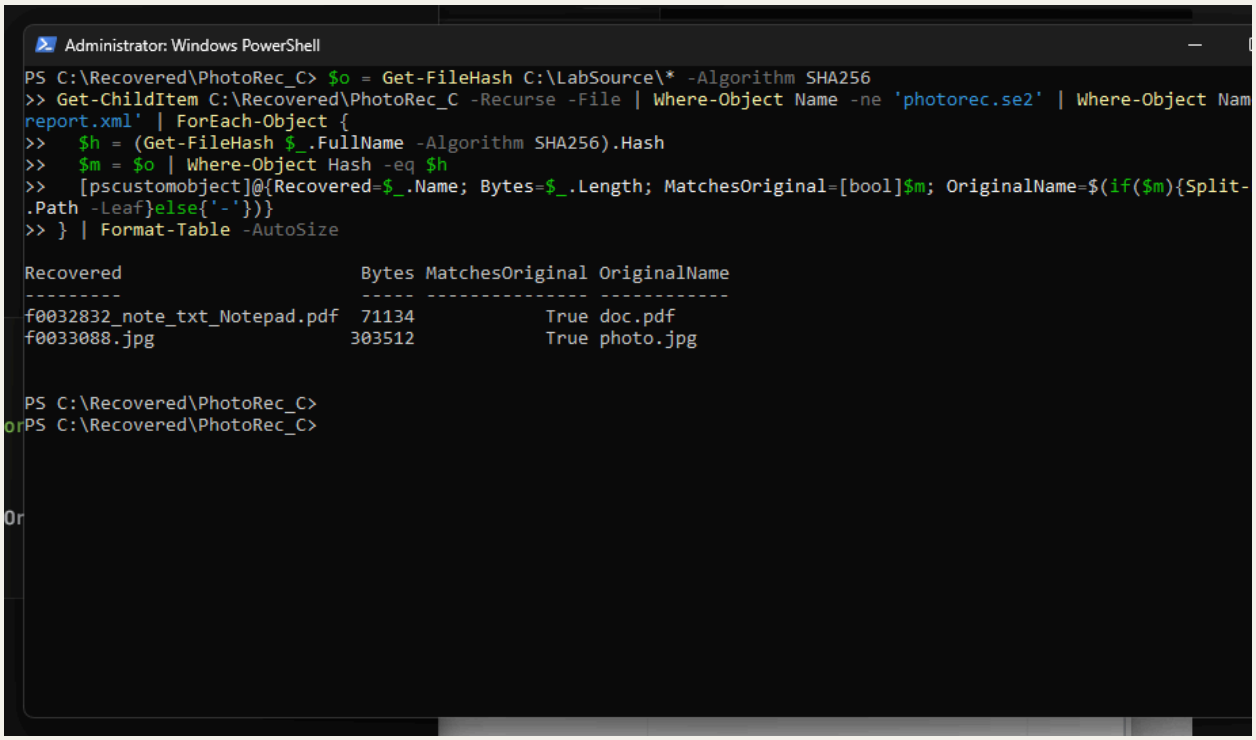


4.8 Recovery of Scenario C with Recuva (Deep Scan)

- Deep Scan took about 14 minutes and found two files with no original names or dates: `[000001].pdf` and `[000002].jpg`.



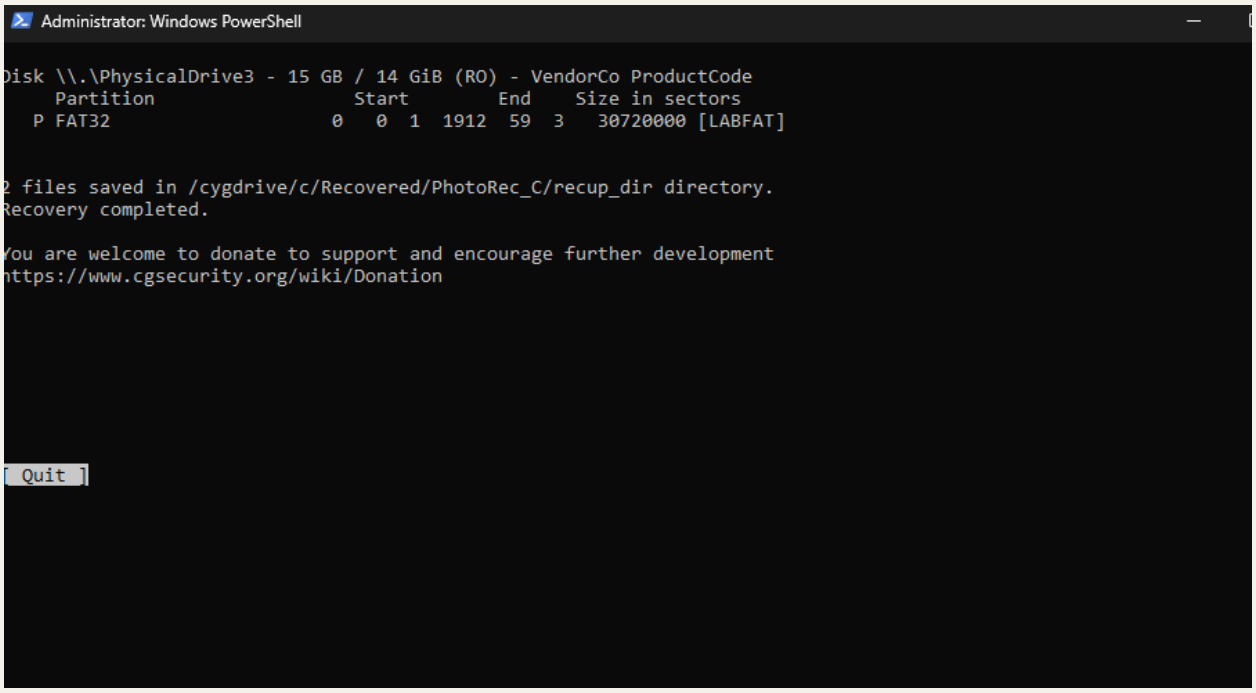
- `[000001].pdf` matched `doc.pdf`. `[000002].jpg` did **not** match by full-file hash.

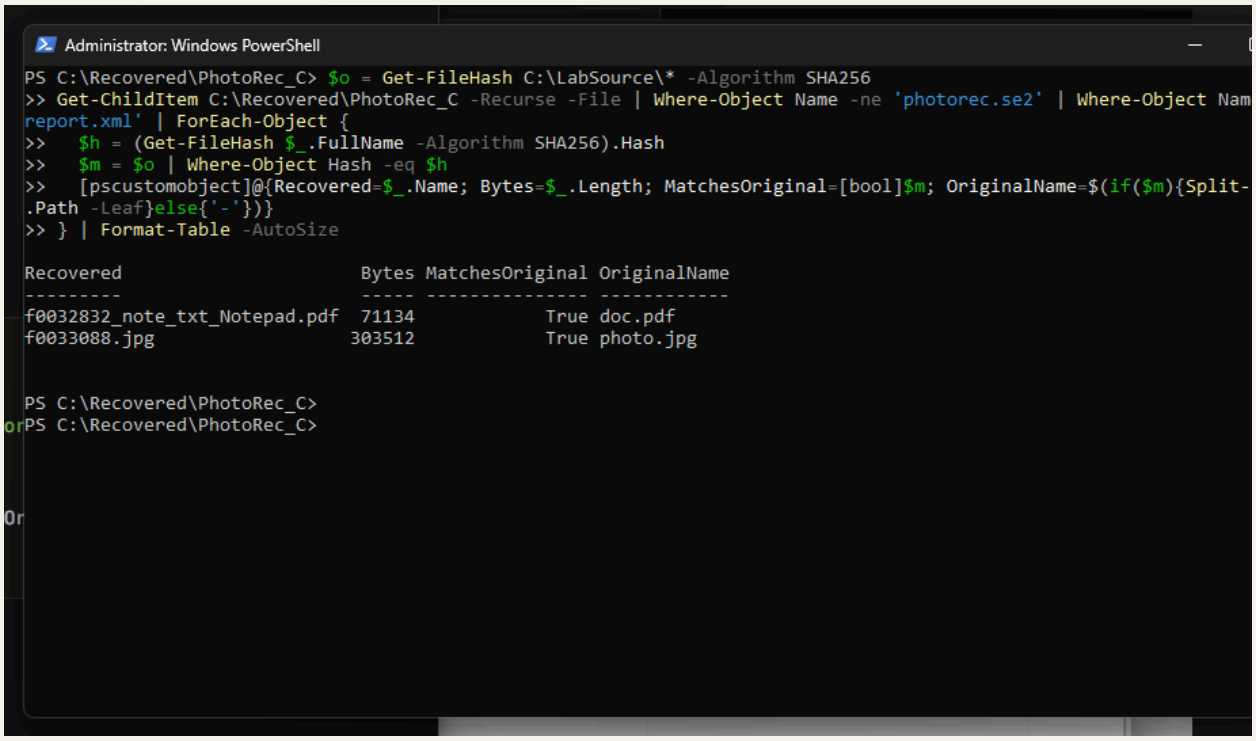


- Diagnosis: the recovered file was 303,616 bytes against 303,512 for the original (104 extra bytes). Hashing only the first 303,512 bytes of the recovered file gave exactly the original hash, so all image data was intact and only padding was added.

4.9 Recovery of Scenario C with PhotoRec

- PhotoRec saved 2 files: `f0032832_note_txt_Notepad.pdf` (71,134 bytes) and `f0033088.jpg` (303,512 bytes). Both matched their originals by hash.


Administrator: Windows PowerShell
Disk \\.\PhysicalDrive3 - 15 GB / 14 GiB (RO) - VendorCo ProductCode
Partition Start End Size in sectors
P FAT32 0 0 1 1912 59 3 30720000 [LABFAT]
2 files saved in /cygdrive/c/Recovered/PhotoRec_C/recup_dir directory.
Recovery completed.
You are welcome to donate to support and encourage further development
https://www.cgsecurity.org/wiki/Donation
[Quit]


Administrator: Windows PowerShell
PS C:\Recovered\PhotoRec_C> $o = Get-FileHash C:\LabSource* -Algorithm SHA256
>> Get-ChildItem C:\Recovered\PhotoRec_C -Recurse -File | Where-Object Name -ne 'photorec.se2' | Where-Object Nam
report.xml' | ForEach-Object {
>> $h = (Get-FileHash $_.FullName -Algorithm SHA256).Hash
>> $m = $o | Where-Object Hash -eq $h
>> [pscustomobject]@{Recovered=$_.Name; Bytes=$_.Length; MatchesOriginal=[bool]$m; OriginalName=$(if($m){Split-
.Path -Leaf}else{'-'})}
>> } | Format-Table -AutoSize
Recovered Bytes MatchesOriginal OriginalName
f0032832_note_txt_Notepad.pdf 71134 True doc.pdf
f0033088.jpg 303512 True photo.jpg
PS C:\Recovered\PhotoRec_C>
PS C:\Recovered\PhotoRec_C>

4.10 Environment evidence

```
PS C:\Recovered\PhotoRec_C>
PS C:\Recovered\PhotoRec_C> Get-CimInstance Win32_OperatingSystem | Select-Object Caption, Version
>> Get-Disk | Where-Object BusType -eq USB | Select-Object FriendlyName, Size, PartitionStyle
>> Get-Item -LiteralPath "C:\LabSource\photo.jpg","C:\Recovered\Recuva_C\[000002].jpg" | Select-Object Name, Leng
>> Get-ChildItem C:\Recovered -Recurse -File | Where-Object Name -notin 'photorec.se2','report.xml' | Select-Obje
'Path';e={$_.FullName.Replace('C:\Recovered\','')}}, Length | Format-Table -AutoSize

Caption                 Version
-------                 -------
Microsoft Windows 11 Pro 10.0.26200




Path                                                Length
----                                                ------
PhotoRec\recup_dir.1\f0032832_note_txt_Notepad.pdf    71134
PhotoRec_C\recup_dir.1\f0032832_note_txt_Notepad.pdf  71134
PhotoRec_C\recup_dir.1\f0033088.jpg                  303512
Recuva\_oc.pdf                                        71134
Recuva\_ote.txt                                       54018
Recuva_C\[000001].pdf                                 71134
Recuva_C\[000002].jpg                                303616
```

5. Results

#	File	Scenario	Tool	Recovered	Hash match	Filename preserved	Opens
1	note.txt	A: normal delete	Recuva	Y	Y	N (`_ote.txt`)	Y
2	note.txt	A: normal delete	PhotoRec	N	n/a	n/a	n/a
3	doc.pdf	B: Shift+Delete	Recuva	Y	Y	N (`_oc.pdf`)	Y
4	doc.pdf	B: Shift+Delete	PhotoRec	Y	Y	N (`f0032832_note_txt_Notepad.pdf`)	Y
5	photo.jpg	C: quick format	Recuva (Deep Scan)	Y	N (content identical, 104 extra padding bytes)	N (`[000002].jpg`)	Y
6	photo.jpg	C: quick format	PhotoRec	Y	Y	N (`f0033088.jpg`)	Y
7	doc.pdf*	C: quick format	Recuva (Deep Scan)	Y	Y	N (`[000001].pdf`)	Y

8	doc.pdf*	C: quick format	PhotoRec	Y	Y	N (`f0032832_note_txt_Notepad.pdf`)	Y
9	note.txt	C: quick format	Recuva (Deep Scan)	N	n/a	n/a	n/a
10	note.txt	C: quick format	PhotoRec	N	n/a	n/a	n/a

* `doc.pdf` had been deleted in Scenario B and its data was still in unallocated space when the stick was quick-formatted.

Summary: 6 of 10 attempts recovered a file, and 5 of those 6 were byte-identical to the originals. The sixth (row 5) contained all of the original data plus 104 bytes of padding.

6. Observations

Why recovery works

On FAT32, deleting a file changes only the metadata. The first byte of the file's directory entry is set to a "deleted" marker (which is why Recuva showed `_ote.txt` and `_oc.pdf`), and the file's clusters are marked free in the FAT. The file's data stays on the disk until the same clusters are reused. Quick format goes one step further by rewriting the boot sector, FAT and root directory, but it does not erase the data area. This explains why `photo.jpg` and `doc.pdf` were recoverable after the format, even though Windows showed an empty drive. Normal delete and Shift+Delete gave the same results, which is consistent with the observation that on a removable drive Windows deletes files immediately instead of using the Recycle Bin.

FAT32 vs NTFS

Only FAT32 was tested in this lab, so the following NTFS points are expected behaviour from how the file systems are designed, not results from this experiment. NTFS keeps a Master File Table record for each file, and a deleted file's record often survives with its name and size intact, so tools like Recuva can often restore original names. FAT32 loses the first character of the name at deletion and, after a quick format, loses the directory entries entirely. Testing NTFS would be a useful extension.

Tool comparison

Aspect	Recuva	PhotoRec
Method	Reads file-system metadata (standard scan) or scans for file signatures (Deep Scan)	Ignores the file system and carves files by internal signatures
Filenames	Partly preserved after a plain delete (first character lost); lost after quick format	Never preserved from the file system; generated names (in one case derived from the PDF's internal title metadata)
Speed in this lab	0.02 s (standard); about 14 min (Deep Scan)	Minutes [FILL IN EXACT TIMES FROM YOUR NOTES]
Plain text recovery	Recovered `note.txt` after a plain delete	Could not recover `note.txt` (no header signature)
File size accuracy	Standard scan exact; Deep Scan padded the JPG to a 512-byte sector boundary	Found the JPG's end marker and recovered it exactly
Interface	Graphical wizard	Text menu, needs care choosing the right disk

When recovery fails

- **Overwriting:** new data written into the freed clusters destroys the content. This is why the stick was left untouched and all output went to C:.
- **No signature:** carving tools cannot identify plain text, and `note.txt` was not recovered by either tool after the quick format, even though its data was likely still on the disk.
- **Fragmentation:** on FAT32 the cluster chain is erased at deletion, so fragmented files are hard to rebuild. The small files here were most likely contiguous.
- **Full format, secure erase or encryption:** a full format that writes zeros, secure-delete tools, or strong encryption make recovery impractical.
- **SSD TRIM:** on TRIM-enabled SSDs the drive itself may erase freed blocks, so recovery is much less likely than on a USB stick.

7. Limitations

- Only FAT32 was tested. NTFS was not tested because of the time available.
- Only three small files were used, and they were probably stored contiguously. Fragmented, larger or many-file cases were not tested.
- The stick was largely empty, so little or no overwriting occurred. Recovery rates would fall on a well-used drive.
- The TestDisk program itself was not used; only PhotoRec (from the same package) was tested.
- One run per scenario and tool, with no repeats, so timings are approximate.
- The explanation for Recuva's 104 extra bytes (sector-aligned padding) is inferred from the size difference (303,616 is a multiple of 512) and the trimmed-hash test, not from Recuva's documentation.
- A USB flash stick behaves differently from an SSD or a mechanical hard disk.

8. Conclusion

On a FAT32 USB stick, files removed by normal delete, Shift+Delete and quick format could be recovered with free tools, and 5 of the 6 recovered files were byte-identical to the originals by SHA-256. The sixth (Recuva Deep Scan, `photo.jpg`) held all the original data plus 104 bytes of padding. Recuva was easier to use and kept part of the filename after a plain delete, while PhotoRec was more exact on file size and did not depend on the file system, but it never preserved names and could not recover a plain text file. The results show that deleting a file or quick-formatting a drive does not securely erase data, and that the best protection against unwanted recovery is overwriting or encryption, while the best

chance of successful recovery is to stop writing to the device immediately and recover to a different drive.

References

- Recuva (Piriform / CCleaner) — https://www.ccleaner.com/recuva
- TestDisk and PhotoRec (CGSecurity) — https://www.cgsecurity.org/wiki/PhotoRec
- Microsoft Learn: Get-FileHash — https://learn.microsoft.com/powershell/module/microsoft.powershell.utility/get-filehash